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Thermally derived evapotranspiration from the Surface Temperature Initiated Closure (STIC) model improves cropland GPP estimates under dry conditions

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A B S T R A C T

Satellite-based gross primary productivity (GPP) monitoring in croplands is challenging due to our limited ability to empirically constrain photosynthetic capacity and associated parameters. Here, we investigated if integrating land surface temperature (T_R)-based evapotranspiration (ET) or latent heat flux (λE) into a Remote sensing-driven approach to Coupling Ecosystem Evapotranspiration and Photosynthesis (RCEEP) model, which is modified from the underlying water use efficiency method, can better characterize GPP under dry conditions as compared to the light use efficiency (LUE)-based models. We developed the new GPP model, termed STIC-RCEEP, by combining an ET model, called STIC (Surface Temperature Initiated Closure), and RCEEP. We compared the performance of STIC-RCEEP against four conventional LUE-based models (Vegetation Photosynthesis Model, MOD17, STIC-MOD17, STIC-LUE), using tower-based daily GPP data from different cropland flux sites across the globe. An evaluation of the five GPP models, all optimized using available 170 site-years data from the 22 flux sites, revealed the relatively better performance of the STIC-RCEEP (R^2 = 0.78 and RMSE = 2.5 gC m^{-2} d^{-1}) with respect to the other four LUE models. Error analysis revealed substantially less error of STIC-RCEEP particularly under the dry conditions and RMSE of STIC-RCEEP was 9%–14% less than the other models. This finding highlights the enhanced ability of thermal sensors to capture the water stress signals in ET and GPP, which is also evidenced by the substantially better performance of STIC than another widely-regarded non-thermal ET model (the Priestly Taylor - Jet Propulsion Laboratory model), under dry conditions. The improved GPP estimates from STIC-RCEEP under water-stressed environment opens avenues for further research and applications using existing and new T_R-based sensors in coupled crop water use-productivity modeling.

1. Introduction

Croplands occupy around 12–14% of the global land area (Yuan et al., 2015; Yuan et al., 2016; Wu et al., 2018), of which over 85% belong to dryland agriculture (Chen et al., 2014). However, sustainable food production in the arid and semi-arid drylands is under serious threat due to climate warming and rainfall anomaly (Lobell et al., 2011; Dai, 2013; Kang and Eltahir, 2018; Marvel et al., 2019; Henchiri et al., 2020). Maximizing crop productivity with limited available water is a sustainable way towards global food security (Godfray et al., 2010; Foley et al., 2011). Agricultural sustainability requires optimization of photosynthesis and transpiration rates to adapt to the impacts of climate change (Parry et al., 2011; Long et al., 2015). Thus, accurate information on crop photosynthesis (referred to as gross primary productivity, GPP, on ecosystem-level) is critical to assess crop yield (van Ittersum et al., 2013; Dusenge et al., 2020). Reasonable characterizations of cropland GPP from regional to global scale under different hydro-climatic settings are crucial to support efforts towards global food security (Beer et al., 2010; Cheng et al., 2017).

Despite the extensive use of remote sensing (RS) methods to

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characterize GPP across multiple biomes (Chen et al., 2014; Kotani et al., 2014; Yuan et al., 2014a; Yuan et al., 2016; Zhang et al., 2016; Lees et al., 2018; Mo et al., 2018; Sun et al., 2019), the potential of regional-scale GPP modeling is hindered due to the lack of a physically-driven method for estimating photosynthetic capacity parameters and the limited ability of the empirical soil and vegetation water stress scalars to constrain GPP (Xiao et al., 2004; Ryu et al., 2011; Wagle et al., 2016; Madani et al., 2017). Although the water balance methods provide a process-based approach to simulate soil and plant water status (Ju et al., 2006; Chen et al., 2012; Bai et al., 2018), their regional implementation is a challenging task due to the lack of irrigation, drainage, and sub-surface soil properties data. Hence, empirical formulations of RS or meteorological water stress factors are widely used in GPP modeling until now (Xiao et al., 2004; Ryu et al., 2011; Wagle et al., 2016; Madani et al., 2017).

The commonly used variables in the empirical formulation of water stress scalars in the contemporary GPP models include atmospheric vapor pressure deficit (VPD) (Chen et al., 1999; Running et al., 1999; Heinsch et al., 2003; Madani et al., 2017) and vegetation indices (VIs) (Xiao et al., 2004; Yebra et al., 2015; Bai et al., 2018) or a combination of VI and thermal infrared (TIR) RS-derived land surface temperature (T_R) (Jiang and Islam, 2001; Wang et al., 2006; Mallick et al., 2009; Garcia et al., 2014; Li et al., 2020). However, these variables may not fully characterize the water stress in vegetation (Mallick et al., 2013; Yuan et al., 2014b; Mallick et al., 2015; Yuan et al., 2015; Bai et al., 2017). For example, VPD can be decoupled from the surface wetness under advective conditions (Yan et al., 2012; Morillas et al., 2013). VIs response to plant water deficit typically lags a few days to weeks (Liu and Kogan, 1996; Liu et al., 2020) and do not explain soil moisture-induced variations in crop productivity (Anderson and Goulden, 2011). The VI (or albedo)-T_R space method estimates surface water status using remotely sensed contextual information of a region of limited size and the method is highly sensitive to the size of the spatial domain (Long et al., 2011; Tang et al., 2013; Bhattarai et al., 2019).

Evapotranspiration (ET) is intrinsically coupled with ecosystem photosynthesis (Beer et al., 2009; Medlyn et al., 2011; Zhou et al., 2014; Zhang et al., 2016) and coupling ET information in the GPP models can effectively capture the water stress-induced variations in crop GPP (Anderson et al., 2011; Anderson et al., 2012). Conventional light use efficiency (LUE) models incorporate ET information empirically either through water balance derived wetness (e.g., Carnegie-Ames-Stanford approach, CASA)) (Potter et al., 1993) or by deriving ET-based water stress scalar from the eddy covariance (EC) measurements (Yuan et al., 2007; Yuan et al., 2016). However, the water stress scalar derived from the EC-LUE approach may not be well representative of crop water status at the regional scale. Recent studies (Yuan et al., 2014a; Zhang et al., 2016; Bai et al., 2021) suggest the underlying water use efficiency (uWUE) methods (Zhou et al., 2014; Zhang et al., 2016) were better than the LUE models in characterizing GPP using ET. In addition, the uWUE approaches can benefit from incorporating remote sensing information (Bai et al., 2021). Hence, the key limitation in existing RS-based GPP models appears to be the characterization of water stress, suggesting that the estimates of cropland GPP under dry conditions could be improved by directly integrating RS-based ET into the GPP models.

ET estimates from T_R-driven surface energy balance (SEB) models can potentially constrain cropland carbon and water fluxes and replace the need for precipitation, irrigation information, and soil moisture modeling as needed by the water balance models (Anderson et al., 2008). This is largely because T_R is very sensitive to soil moisture variations and evaporative cooling, and carries the imprints of vegetation water stress (Anderson et al., 2008; Kalma et al., 2008; Anderson et al., 2012). Several T_R-driven SEB models, such as the two source energy balance (TSEB) model (Kustas and Norman, 1996; Kustas and Norman, 1999), Surface Temperature Initiated Closure (STIC) model (Mallick et al., 2014b; Mallick et al., 2015; Bhattarai et al., 2018; Mallick et al., 2018b), Surface Energy Balance System (SEBS) (Su, 2002; Chen et al., 2013), and other contextual based models (Bastiaanssen et al., 1998; Allen et al., 2007) have been developed and validated globally. Among these SEB models, STIC derives ET without the need for any empirical formulations of the biophysical conductances (i.e., aerodynamic, g_a and canopy conductance, g_c, respectively) and the aerodynamic temperature (T_0) (Mallick et al., 2014b; Mallick et al., 2015; Mallick et al., 2016; Mallick et al., 2018b). The intercomparison of STIC with several contemporary SEB models revealed comparable or better performances of STIC particularly in the arid and semi-arid ecosystems of the US (Bhattarai et al., 2018), Australia (Trebs et al., 2021), and India (Bhattarai et al., 2019).

In this study, we integrated STIC and the RS-driven approach to Coupling Ecosystem Evapotranspiration and Photosynthesis (RCEEP) (Bai et al., 2021), modified from the uWUE method (Zhou et al., 2014), to develop a new GPP model, called STIC-RCEEP. We tested the ability of STIC-RCEEP to estimate daily cropland GPP across a broad spectrum of aridity representing different dryness conditions. We hypothesize that STIC-RCEEP is relatively more efficient as compared to the conventional LUE models in estimating GPP and the integration with STIC ET enables the model to capture the effects of water stress in GPP particularly in the dry croplands. The specific objectives of the present study are as follows:

(1) Integrate STIC-derived ET into RCEEP formulation to develop STIC-RCEEP model,
(2) Calibrate and validate the STIC-RCEEP model to simulate daily GPP from multiple global cropland flux sites covering a wide range of environmental conditions,
(3) Compare the performance of STIC-RCEEP against other four conventional LUE models for a broad spectrum of moisture conditions;
(4) Analyze the impact of errors in the input variables and estimated ET on the performance of STIC-RCEEP.

2. Data and methods

2.1. Data and data preprocessing

2.1.1. Flux-site data

Energy, water, and carbon flux observations from a total of 22 cropland flux sites (Table 1 and Table C1), sampling a wide range in climate conditions, were used in the present study. These sites belong to three widely used flux networks, namely, FLUXNET2015 (FLUXNET, https://fluxnet.fluxdata.org/), European Flux Database Cluster (EFDC, http://www.europe-fluxdata.eu/), and AmeriFlux (https://ameriflux.lbl.gov/). Sites in orchards, unknown crop species, or lacking at least one year of GPP and latent heat flux (λE) observations were excluded from this study. Several GPP versions are available in the FLUXNET sites, from which we used the version labeled as 'GPP_NT_VUT_REF'. In this version, 'NT' indicates that the GPP was retrieved by partitioning the observed net ecosystem carbon exchange (NEE) using the nighttime method (Reichstein et al., 2005), 'VUT' denotes the data was filtered using the friction velocity threshold (https://fluxnet.org/data/fluxnet2015-dataset/data-processing/), and 'REF' denotes the most common reference value among those from 40 different FLUXNET2015 work-groups (Bai et al., 2018). The EFDC provides four nighttime method-derived GPP versions with no noticeable differences in their outputs, among which we used the one labeled as 'GPP_st_MDS' for further analysis. NEE in this GPP version is calculated using the storage obtained with the discrete approach (a single point on the top of the tower) with the same method for all the EFDC sites, and the NEE gaps were filled using the Marginal Distribution Sampling method (Reichstein et al., 2005). It should be noted that AmeriFlux does not provide uniform methods to generate GPP from NEE and available GPP values in AmeriFlux sites are provided by the principal investigators of the sites.

Table 1
An overview of 22 cropland flux sites used in the present study.

Site code	Site name	Crop type	Irrigation [a]	Country	Latitude	Longitude	Network	Citation
US-Tw3	Twitchell Alfalfa	Alfalfa	Yes	USA	38.1159	−121.6467	FLUXNET	Baldocchi et al. (2015)
US-Twt	Twitchell Island	Rice	Yes	USA	38.1087	−121.6530	FLUXNET	Hatala et al. (2012)
US-Tw2	Twitchell Corn	Maize	Yes	USA	38.1047	−121.6433	FLUXNET	Knox et al. (2014)
ES-ES2	El Saler-Sueca	Rice	Yes	Spain	39.2755	−0.3152	EFDC	Carvalhais et al. (2010)
US-ARM	ARM Southern Great Plains site–Lamont	Wheat/Maize/Soybean/Canola/Cowpeas	No	USA	36.6058	−97.4888	FLUXNET	Raz-Yaseef et al. (2015)
FR-Lam	Lamasquere	Triticale/Maize/Wheat	Yes	France	43.4965	1.2379	EFDC	Béziat et al. (2009)
FR-Aur	Aurade	Rapeseed/Wheat/Sunflower	No	France	43.5496	1.1062	EFDC	Béziat et al. (2009)
US-Ne3	Mead–rainfed maize–soybean rotation site	Maize/Soybean	No	USA	41.1797	−96.4397	FLUXNET	Suyker and Verma (2012)
DE-RuS	Selhausen Juelich	Sugar beet/Winter cereals/Potato	-	Germany	50.8659	6.4472	FLUXNET	Eder et al. (2015)
US-CRT	Curtice Walter–Berger cropland	Soybean/Wheat	-	USA	41.6285	−83.3471	FLUXNET	Chu et al. (2014)
DE-Geb	Gebesee	Wheat/Potato/Sugar beet	No	Germany	51.1001	10.9143	FLUXNET	Anthoni et al. (2004)
US-Ne1	Mead–irrigated continuous maize site	Maize	Yes	USA	41.1651	−96.4766	FLUXNET	Verma et al. (2005)
FR-Gri	Grignon	Barley/Mustard/Wheat/Maize/ Triticale	-	France	48.8442	1.9519	FLUXNET	Loubet et al. (2011)
US-Ne2	Mead–irrigated maize–soybean rotation site	Maize/Soybean	Yes	USA	41.1649	−96.4701	FLUXNET	Suyker and Verma (2012)
US-Ro1	MN - Rosemount- G21	Maize/Soybean	-	USA	44.7143	−93.0898	AmeriFlux	Bavin et al. (2009)
DE-Seh	Selhausen	Sugar beet/Wheat/Maize	-	Germany	50.8706	6.4497	FLUXNET	Korres et al. (2013)
IT-Cas	Castellaro	Rice	Yes	Italy	45.0628	8.6685	EFDC	Meijide et al. (2011)
BE-Lon	Lonzee	Wheat/Mustard/Potato/Sugar beet	No	Belgium	50.5516	4.7461	FLUXNET	
DE-Kli	Klingenberg	Barley/Rapeseed/Maize/Wheat	Yes	Germany	50.8929	13.5225	FLUXNET	Brust et al. (2012)
IT-BCi	Borgo Cioffi	Maize/Fennel/Clover/Ryegrass/ Alfalfa	Yes	Italy	40.5238	14.9574	FLUXNET	Ranucci et al. (2010)
DK-Ris	Risbyholm	C_3 crops	Yes	Denmark	55.5303	12.0972	EFDC	Kutsch et al. (2010)
CH-Oe2	Oensingen2 crop	Wheat/Barley /Potato/Rapeseed	No	Switzerland	47.2863	7.7343	FLUXNET	Moors et al. (2010)

[a] The short dash "-" indicates the irrigation information was unclear.

2.1.2. Dryness of the site-year

For each site-year, we computed the dryness (Table C1) of the growing season in all the flux sites and subsequently assessed the performance of ET and GPP models under different dryness scenarios. The dryness of each site-year was computed using the following steps:

(a) *Define the growing season*: Successive 4-month having the largest GPP was defined as the growing season. If the identified growing season period does not cover more than 90 days with available flux-site GPP values or if there are no available GPP values of this site-year, GPP in this step was substituted by the solar radiation intercepted by vegetation canopy ($R_g \times f_c$; where R_g and f_c denote the solar radiation and faction vegetation cover, respectively).
(b) *Compute the dryness of each site-year*:

$$dryness = \frac{P_{\mathrm{GS}-16}}{\mathrm{ET}_{\mathrm{P,GS}}} \tag{1}$$

where $\mathrm{ET}_{\mathrm{P,GS}}$ denotes the growing season potential ET; subscript "GS" indicates the growing season period defined in step (a); $P_{\text{GS-16}}$ denotes the cumulative precipitation during the period that starts 16 days earlier than does the GS and has the same length with GS. Considering the water storage capacity of the soil, we used $P_{\text{GS-16}}$ rather than P_{GS} in Eq. (1) to represent the natural water supply during the growing season. ET_{P} here was calculated using the Priestley-Taylor equation (Priestley and Taylor, 1972) due to the difficulty in quantifying the aerodynamic conductance in the Penman-Monteith equation (Monteith, 1995).

It is not reasonable to use annual P and ET_{P} to characterize the dryness of the site-year in some cases. For example, in the Mediterranean climate with hot dry summer and cold wet winter, the ratio of annual P to ET_{P} could fail to reflect the dryness severity during summer which is also the peak growing season of crops. Given our focus on crop photosynthesis in this study, site-year dryness defined as the dryness condition of the crop growing season (Eq. (1)) is more reasonable.

2.1.3. MODIS remote sensing data and ERA-5 meteorological data

To run the ET and GPP models, we utilized remotely sensed data from several MODIS products including (1) 1-km daytime and nighttime land surface temperature (T_{R}) data on a daily step from the MOD11A1 product (for driving ET estimates from STIC and GPP estimates from STIC-RCEEP); (2) 500-m, daily black-sky and white-sky albedo for shortwave broadband from MOD43A3 (for computing net surface radiation, R_{n}); (3) 500-m, 8-day surface reflectance data from MOD09A1 (for computing Land Surface Water Index, LSWI); and (4) 250-m, 16-day normalized vegetation index (NDVI) and enhanced vegetation index (EVI) from the MOD13Q1 and MYD13Q1 (for computing the fractional vegetation cover (f_c) and driving the new GPP model, respectively). In this study, We computed f_c following the quadratic method (Carlson and Ripley, 1997).

$$f_c = \left(\frac{\mathrm{NDVI} - \mathrm{NDVI_S}}{\mathrm{NDVI_V} - \mathrm{NDVI_S}}\right)^2 \tag{2}$$

where $\mathrm{NDVI_S}$ and $\mathrm{NDVI_V}$ denote the NDVI values for bare soil and fully covered vegetation, respectively. The values of $\mathrm{NDVI_S}$ and $\mathrm{NDVI_V}$ depend on vegetation and soil types and wetness (Gao et al., 2020), which can differ between different years for a specific site, thus we estimated $NDVI_S$ and $NDVI_V$ for each site-year as 95th and 5th percentile of the daily NDVI values in the corresponding site-year.

Daily meteorological data for running the ET and GPP models were retrieved from the ERA-5 reanalysis meteorological dataset (https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5). The ERA-5 dataset provides gridded meteorological variables with a spatial resolution of 0.125 arc-degree on an hourly step. Four variables including R_g, wind speed (u), air temperature (T_a), and dew point (T_{dew}) were used in this study. T_{dew} was used in combination with T_a to compute the vapor pressure deficit (VPD) and relative humidity (RH) (Allen et al., 1998). hourly values of these variables were aggregated to derive corresponding daily values. Instantaneous net radiation (R_n) for driving STIC was computed from the hourly R_g at the satellite overpass time in conjunction with surface albedo, T_a, RH, and MODIS T_R (Appendix A). Daily R_n was computed from daily meteorological variables using the method outlined by Allen et al. (1998).

2.2. The STIC-RCEEP model

Transpiration (ET_T) is intrinsically coupled with photosynthesis through the stomatal conductance. Since ET_T is affected by the soil water content variability, it also captures the effects of water stress on photosynthesis. In addition, ET_T is a dominating component of ET (Jasechko et al., 2013). Therefore, cropland GPP estimation can be improved by incorporating the information of ET into the GPP model. To accomplish this, we integrated the T_R-based SEB modeling approach with a relatively new model, RCEEP (Bai et al., 2021), which represents a biophysical linkage between GPP and ET. RCEEP provides a straightforward approach for estimating GPP from ET. The original framework of RCEEP was modified to compute cropland GPP from the T_R-based ET and ET_T. ET_T was estimated by partitioning T_R-driven ET from the STIC model (Mallick et al., 2014b; Mallick et al., 2015; Mallick et al., 2016; Mallick et al., 2018a) and partitioning of ET to estimate ET_T was done following Fisher et al. (2008) and Yao et al. (2013). We named this new model STIC-RCEEP.

2.2.1. Description of STIC-RCEEP model

RCEEP (Bai et al., 2021) was modified from the underlying water use efficiency (uWUE) approach (Zhou et al., 2014) to better characterize the relationship between ecosystem GPP and ET. The uWUE is a generic method that represents the biophysical coupling between GPP and ET, and it uses leaf-to-air vapor pressure difference (D) to account for the stomata responses to transpiration (Zhou et al., 2014, 2015). Since D is difficult to obtain on the regional scale, VPD was used as a surrogate of D in the actual uWUE method. This is, however, prone to errors due to the significant differences between the air and leaf surface temperature under water-stressed conditions, which very often prevail in the arid and semi-arid ecosystems (Jackson et al., 1981; Idso et al., 1982; Almeida, 1986; Olufayo et al., 1993; Nelson and Bugbee, 2015). To better characterize the relationship between ET and GPP, STIC-RCEEP partitions ET_T from ET following Fisher et al. (2008) and Yao et al. (2013), and couples ET_T and GPP using RS-based stomatal conductance instead of D (Bai et al., 2021). The specific ET-GPP coupling part in the STIC-RCEEP model has already been validated across multiple biomes at a global scale (Bai et al., 2021). The analytical expressions of ET-GPP coupling approach in the STIC-RCEEP model are presented as follows.

$$\mathrm{GPP}=\begin{cases}\left(\mathrm{uWUE}\cdot\sqrt{b}\right)\times\sqrt{\dfrac{1.3\mathrm{EVI}\cdot\mathrm{ET}^{s}}{P_a\cdot\mathrm{ET}^{s}_{max}}\times\mathrm{ET_T}} & ,T_a>0\\ 0 & ,T_a\leq 0\end{cases}\tag{3}$$

$$\mathrm{ET_T}=\mathrm{ET}\times F_t\tag{4}$$

$$F_t=max\left(\frac{f_g f_T f_c}{f_{SM}+f_c\left(f_g f_T-f_{SM}\right)},f_c\right)\tag{5}$$

where both ET and ET_T are measured in mol m^{-2} s^{-1}; uWUE•$\sqrt{b}$ is a single quantity measured in μmol C (mol H_2O)$^{-1}$ kPa$^{0.5}$, associated with the water use efficiency of the ecosystem; P_a denotes the atmosphere pressure (kPa); s an empirical parameter (0.31); ET_{max} the maximum value of ET observed under the natural environment (6.8×10^{-3} mol m^{-2} s^{-1}); F_t the fraction of ET_T in ET; f_g, f_T, and f_{SM} the RS-based empirical estimates of the fraction of green canopy, temperature-constraint on ET_T, and soil moisture.

Here, we integrated the T_R-driven SEB model, called STIC (Surface Temperature Initiated Closure) (Mallick et al., 2014b; Mallick et al., 2015; Mallick et al., 2016), with RCEEP to develop a new GPP model, termed STIC-RCEEP. In this new model, the ET term in Eqs. (3) and (4) was substituted for the T_R-derived ET from the STIC model to compute daily GPP. The three factors, f_g, f_T, and f_{SM}, were computed following Bai et al. (2021). The STIC model has been widely evaluated across a wide range of biomes (from tropical rainforest to woody savanna) (Mallick et al., 2014b), wetland, and aridity gradients in South America (Mallick et al., 2016), different ecological transects in Australia (Mallick et al., 2018a), US (Mallick et al., 2014b; Mallick et al., 2015; Bhattarai et al., 2018), and India (Bhattarai et al., 2019). The description of the STIC model is given below.

2.2.2. An overview of STIC

STIC is a one-dimensional analytical non-parametric SEB model and it is based on an integration of satellite T_R observations into the Penman-Monteith Energy Balance (PMEB) equation (Monteith, 1965). One of the fundamental assumptions in STIC is the first-order dependence of the two critical conductances of the PMEB equation on a water stress factor (M). In STIC, this water stress factor is physically retrieved through T_R (Mallick et al., 2014b; Mallick et al., 2015; Mallick et al., 2016; Mallick et al., 2018a; Mallick et al., 2018b; Trebs et al., 2021). It is hypothesized that linking M with the conductances will simultaneously integrate the information of T_R into the PMEB model. The common expressions for λE and H according to the PMEB equation are as follows:

$$\lambda\mathrm{E}=\frac{\Delta\cdot\phi+\rho_a\cdot c_P\cdot g_a\cdot\mathrm{VPD}}{\Delta+\gamma(1+g_a/g_c)}\tag{6}$$

$$H=\frac{\gamma\cdot\phi\cdot(1+g_a/g_c)-\rho_a\cdot c_P\cdot g_a\cdot\mathrm{VPD}}{\Delta+\gamma(1+g_a/g_c)}\tag{7}$$

where Δ is the gradient of the saturated vapor pressure to T_a; ϕ is the available energy; g_a and g_c are the aerodynamic conductance and canopy-substrate conductance, respectively; c_P is the specific heat of air at constant pressure; γ is the psychrometric constant; and ρ_a is the air density.

In the above equations, g_a and g_c are unknown and the STIC methodology is based on finding analytical solutions for the two unknown conductances to directly estimate λE (Mallick et al., 2016; Mallick et al., 2018a). The need for such analytical estimation of these conductances is motivated by the fact that g_a and g_c can neither be measured at the canopy nor at larger spatial scales, and there are no universally agreed appropriate models of g_a and g_c that currently exist (Matheny et al., 2014; van Dijk et al., 2015). By integrating T_R with standard SEB theory and vegetation biophysical principles, STIC formulates multiple state equations to eliminate the need of using the empirical parameterizations of the g_a, g_c, and the aerodynamic temperature (T_0) as well. It simultaneously bypasses the scaling uncertainties in regional-scale ET mapping due to improper characterizations of the T_0, g_a, and g_c. The state equations for the conductances and T_0 are expressed as a function of those variables that are mostly available as RS observations (e.g., T_R) and weather forecasting models. The four equations are based on standard surface energy balance theory, advection-aridity hypothesis (Brutsaert and Stricker, 1979), and Bowen ratio (β) theory (Bowen, 1926) where β is expressed in terms of the evaporative fraction (Λ) (Shuttleworth et al., 1989; Mallick et al., 2014b; Mallick et al., 2015; Mallick et al., 2016; Bhattarai et al., 2018). The four states equations of STIC are as follows and their detailed derivations are given in Mallick et al. (2016).

$$g_a = \frac{R_n - G}{\rho_a \cdot c_P \cdot ((T_0 - T_a) + (e_0 - e_a)/\gamma)} \tag{8}$$

$$g_c = g_a \cdot \frac{e_0 - e_a}{e_0^* - e_0} \tag{9}$$

$$T_0 = T_a + \frac{e_0 - e_a}{\gamma \cdot (1 - \Lambda)/\Lambda} \tag{10}$$

$$\Lambda = \frac{2\alpha \cdot \Delta}{2\Delta + 2\gamma + \gamma \cdot g_a/g_c \cdot (1 + M)} \tag{11}$$

where e_0*, e_0, and e_a are the saturation vapor pressure and atmospheric vapor pressure at the source-sink height, and atmospheric vapor pressure at the height of T_a measurement, respectively; and α is the Priestley-Taylor coefficient (Priestley and Taylor, 1972). In the state equations, a direct connection to T_R is established by estimating M as a function of T_R (Fig. 1). The information of M is subsequently used in the state equations of conductances, aerodynamic variables (aerodynamic temperature, aerodynamic vapor pressure), and evaporative fraction, which is eventually propagated into their analytical solutions. M is a unitless quantity, which describes the relative wetness (or dryness) of a surface and also controls the transition from potential to actual evaporation, which implies $M \to 1$ under saturated surface conditions and $M \to 0$ under extremely dry conditions. Since T_R is extremely sensitive to the surface moisture variations, it is extensively used for estimating M in a physical retrieval scheme (detail in Appendix A3 of Mallick et al. (2016), Mallick et al. (2018a), and Bhattarai et al. (2018)).

Given the estimates of M, R_n, G, T_a, and RH or e_a, the four state equations (Eqs. (8)–(11)) can be solved simultaneously to derive analytical solutions for the four unobserved variables and to estimate λE through the PMEB model where λE estimates are independent of empirical parameterizations for both g_a and g_c. However, the analytical solutions to the four state equations contain three accompanying unknowns, e_0, e_0*, and α, and as a result, there are four equations with seven unknowns. Consequently, an iterative solution and additional closure equations were needed to determine the three unknown variables (see Fig. 1 or Appendix A2 in Mallick et al. (2016), Mallick et al. (2018a), and Bhattarai et al. (2018)). Once the analytical solutions of g_a and g_c are obtained, both variables are returned into the PMEB equation to directly estimate λE.

The inputs that are directly used in the derivation of λE in STIC include T_a, T_R, RH, R_g, and surface albedo. A short description of these input variables and their use in the derivation of other intermediate variables including their variation, uncertainties, and sources are listed in Table 2. Note that a detailed analysis of how uncertainties in input variables affect ET estimates from STIC is beyond the scope of this study, as the model has been widely evaluated across a wide range of aridity gradients and biomes, as we mentioned in previous contents.

2.2.3. Upscaling the instantaneous ET from STIC to a daily scale

Given daytime T_R retrievals from the MOD11A1 product are available once per day for a single snapshot during the equatorial crossing time of MODIS, the retrievals of ET were instantaneous and instantaneous ET needs to be upscaled for estimating daily ET. The conventional method assumes diurnal conservation of evaporative fraction (Λ) and multiplies instantaneous Λ with daily R_n to derive daily ET. Such a method could be prone to errors because Λ can change throughout the day. In this study, we refer to Bhattarai et al. (2019) and Tang et al. (2017), and upscale the instantaneous ET from STIC to a daily scale based on the reference ET (ET_0) method.

$$ET_{daily} = ETrF \times ET_{0,daily} \tag{12}$$

$$ETrF = \frac{ET_{inst}}{ET_{0,inst}} \tag{13}$$

where ET_{inst} and ET_{daily} denote the actual ET at instantaneous and daily scale, respectively; ETrF is the ratio of actual ET to ET_0; and $ET_{0,inst}$ and $ET_{0,daily}$ the reference ET of the same instance and day. $ET_{0,inst}$ and $ET_{0,daily}$ were computed using the standard ASCE-PM equation (Walter et al., 2001). On rainy or cloudy days, ETrF could be missing due to the unavailability of T_R. These gaps were filled by linearly interpolating the nearest available data in time.

There are several other methods proposed for upscaling ET_{inst} into ET_{daily} (Wandera et al., 2017) and a comprehensive evaluation of these

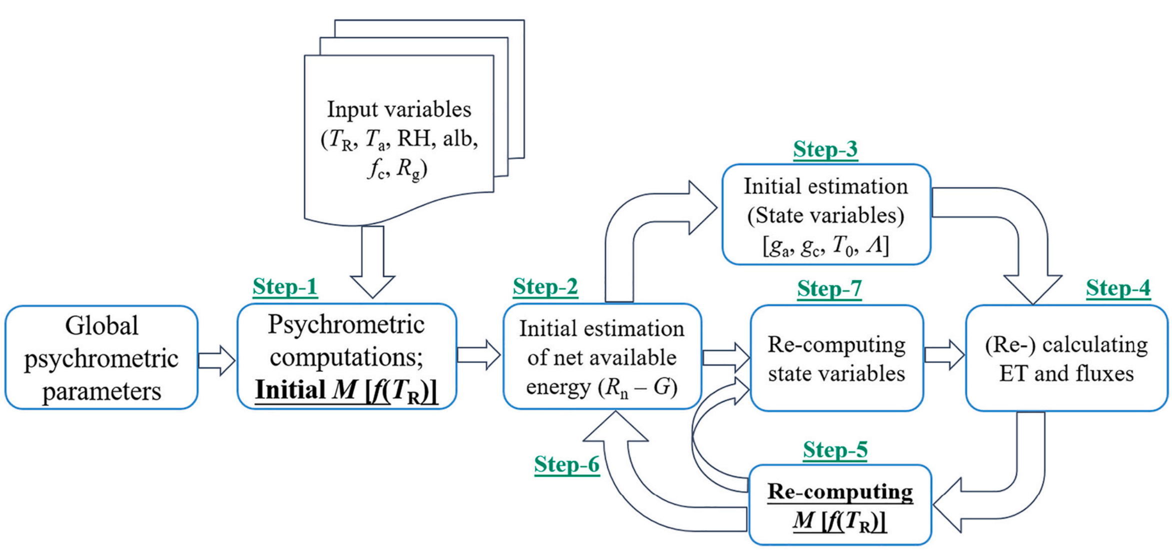

Fig. 1. Methodological flowchart of the STIC model. The input variables T_R, T_a, RH, alb, f_c, and R_g denote the thermal infrared-derived land surface temperature, air temperature, air relative humidity, surface albedo, and global solar radiation, respectively. M denotes the surface wetness and M[f(T_R)] indicates that M is derived from T_R. R_n and G denote the net radiation and ground heat flux, respectively. g_a, g_c, T_0, and Λ denote the aerodynamic conductance, canopy-substrate conductance, aerodynamic temperature, and evaporation fraction, respectively.

Table 2
Description of key input and derived variables in the STIC model with their range, uncertainties (only for direct inputs), and sources.

Type	Variable	Description (unit)	Range	Uncertainties	Sources
Direct input	T_R	Land surface temperature (K)	273–333	±3 K	MOD11A1
	T_a	Air temperature (°C)	0–50	±2 K	ERA-5
	RH	Relative humidity (%)	0–100	±10%	ERA-5
	R_g	Shortwave radiation (W m^{-2})	0–1300	±10%	ERA-5
	alb	Surface albedo	0–1	±10%	MCD43A3
	NDVI	Normalized difference vegetation index	0–1	±10%	MOD13Q1 and MYD13Q1
Derived	γ	Psychrometric constant (hPa $°C^{-1}$)	0.66–0.69	–	Derived from T_a
	ρ_a	Air density (kg/m3)		–	Derived from T_a
	c_P	Specific heat of air at constant pressure (J kg^{-1} $°C^{-1}$)	Fixed to 1005 J kg^{-1}	–	–
	VPD	Vapor pressure deficit of air (hPa)	0–60	–	Derived from T_a and RH
	Δ	Gradient of the saturated vapor pressure to T_a (hPa $°C^{-1}$)	0.04–6.13	–	Derived from T_a
	e_a	Atmospheric vapor pressure at the height of T_a measurement (hPa)	0–120	–	Derived from T_a and VPD
	R_n	Net radiation (W m^{-2})	0–800	–	Derived from R_g, T_R, alb, T_a
	G	Ground heat flux (W m^{-2})	0–200	–	Iterative updating
	g_a	Aerodynamic conductance (m s^{-1})	0–0.1	–	Iterative updating
	g_c	Canopy conductance (m s^{-1})	0–0.1	–	Iterative updating
	T_0	Aerodynamic temperature (°C)	$T_R \pm 15$ K	–	Iterative updating
	Λ	Evaporative fraction	0–1	–	Iterative updating
	e_0	Atmospheric vapor pressure at the source-sink height (hPa)	0–100	–	Iterative updating
	e_0*	Saturation vapor pressure source-sink height (hPa)	3–200	–	Iterative updating
	M	Surface moisture availability (unitless)	0–1	–	Iterative updating involving T_R and T_a
	α	Priestley-Taylor coefficient (unitless)	0–2	–	Iterative updating

methods is beyond the scope of this study. However, when we compared the method (Eq. (12)) used in this study with other two methods, one based on solar radiation and relative humidity to account for the diurnal variations in evaporative fraction (Hoedjes et al., 2008) and based on exo-atmospheric solar radiation (Ryu et al., 2012), we found no noticeable difference in the results (results not shown). We believe that any small differences in ET_{daily} from different upscaling approaches are nullified during the model optimization process (described later in Section 2.4.1) and hence the selection of the upscaling approach will not change the main findings of this paper.

2.3. Four light use efficiency models for comparisons

To test the efficacy of STIC-RCEEP, we evaluated other four light use efficiency (LUE) models, namely the vegetation photosynthesis model (VPM) (Xiao et al., 2004), The MODIS GPP algorithm (MOD17) (Heinsch et al., 2003), a modified version of MOD17 computing the water stress factor (W_S) using the CASA W_S formulation (Potter et al., 1993) along with STIC ET (STIC-MOD17), and a modified version of the EC-LUE model (Yuan et al., 2007; Yuan et al., 2016) driven by STIC ET (STIC-LUE) (Table 3). While VPM and the MOD17 algorithm uses W_S from the remotely sensed LSWI and the meteorological variable VPD, respectively, STIC-MOD17 and SITC-LUE use STIC ET to empirically drive W_S. These LUE models estimate GPP following a simple framework that involves absorbed photosynthetic active radiation (APAR) and optimum LUE as follows.

$$GPP = \varepsilon_0 \cdot f \cdot APAR \tag{14}$$

where ε_0 denotes the optimum of LUE (gC MJ^{-1}); f is the environmental constraints on LUE. The calculations of f differ between different models and are shown in Table 3. APAR for the four LUE models was computed as follows.

$$APAR = 0.5 R_g \cdot f_c \cdot Sec_{daily} \tag{15}$$

where Sec_{daily} denotes the length of a day measured in seconds (86,400 s); f_c the fractional cover of vegetation; R_g the global solar radiation. The number 0.5 in Eq. (15) is an empirical estimate of the proportion of photosynthetic active radiation (PAR) in R_g.

Table 3
The calculation of the environmental scalars (f) of four LUE models used in the present study.[a]

LUE model	Calculation of f	Descriptions of the variables
VPM	$f = f_T \cdot W_S \cdot P_S$ $f_T = \frac{(T_a - T_{min})(T_a - T_{max})}{(T_a - T_{min})(T_a - T_{max}) - (T_a - T_{opt})^2}$ $W_S = (1 + LSWI)/1 + LSWI_{max}$ $P_S = (1 + LSWI)/2$	f_T—Temperature scalar; W_S—Water stress scalar; P_S—Scalar accounting for the effect of leaf age; LSWI—Remotely Sensed Land Surface Water Index; $LSWI_{max}$—The maximum value of LSWI in each site-year;
MOD17	$f = f_T \cdot W_S$ $W_S = \begin{cases} 0.1 & , VPD > VPD_{max} \\ \frac{VPD_{max} - VPD}{VPD_{max} - VPD_{min}} & , VPD_{min} \le VPD \le VPD_{max} \\ 1.0 & , VPD < VPD_{min} \end{cases}$	VPD—Vapor pressure deficit; VPD_{min}—The minimum value of VPD for stomata open; VPD_{max}—The maximum value of VPD for stomata open; λE—Latent heat flux; λE_P—Latent heat flux equivalent to potential ET; R_n—Net radiation.
STIC-MOD17	$f = f_T \cdot W_S$ $W_S = 0.5 + 0.5(\lambda E)/(\lambda E_P)$	
STIC-LUE	$f = \min(f_T, W_S)$ $W_S = (\lambda E)/R_n$	

[a] The calculations of f_T in MOD17, STIC-MOD17, and STIC-LUE were the same as that in VPM. T_{max}, T_{opt}, and T_{min} values are referred from Bai et al. (2018) as 44.6 °C, 23.4 °C, and −2.5 °C for C_3 crops and 43.7 °C, 31.3 °C, and 7.5 °C for C_4 crops, respectively. The values of VPD_{max} and VPD_{min} were not differentiated between C_3 and C_4 crops in MOD17 (Heinsch et al., 2003), as 41.0 hPa and 6.5 hPa of VPD_{max} and VPD_{min} were applied for all crops. The daily λE from STIC was used to compute the W_S in STIC-MOD17 and STIC-LUE. λE_P was computed using the Penman equation along with the aerodynamic conductance and ground heat flux derived from STIC.

Table 4
Optimal parameters of four GPP models.

Model	C3		C4	
	ε_0 (gC MJ^{-1})	uWUE•$\sqrt{b}$ (μmol C (mol H_2O)$^{-1}$ kPa$^{0.5}$)	ε_0 (gC MJ^{-1})	uWUE•$\sqrt{b}$ (μmol C (mol H_2O)$^{-1}$ kPa$^{0.5}$)
VPM	2.010	–	2.974	–
MOD17	1.444	–	2.272	–
STIC-MOD17	1.602	–	2.220	–
STIC-LUE	1.926	–	2.227	–
STIC-RCEEP	–	3264	–	4016

2.4. Model optimizations and evaluations

2.4.1. Model optimizations

The parameters uWUE•$\sqrt{b}$ in STIC-RCEEP and ε_0 in the four LUE models (VPM, MOD17, STIC-MOD17, and STIC-LUE) were optimized prior to model implementation and evaluation. We specifically calibrated the models using the tower-derived (referred to as 'observed' hereafter) GPP and ET. Given STIC is a non-parametric model, no parameter calibration in STIC was required. As only limited numbers of flux sites were available in this study, we did not divide the dataset into calibration and validation subsets, as the primary objective was to compare the effectiveness of the T_R-driven coupled model, STIC-RCEEP, against other four LUE models regarding estimating GPP. The information about all flux sites is presented in Table 1 and the availability of the site-years of these sites is presented in Appendix C. Other essential parameters required for running these models were retrieved from the existing studies. The value of ε_0 is model specified, depending on the factors that are considered in the LUE model and the way these factors are calculated. We optimized uWUE•$\sqrt{b}$ for STIC-RCEEP and ε_0 for each of the four LUE models, separately for C_3 and C_4 crops, using the least-squares method along with observed GPP and ET on a daily scale (see Table 4).

2.4.2. Model evaluation

Commonly used metrics, such as coefficient of determination (R^2), root mean square error (RMSE), and mean bias (Moriasi et al., 2007), were used to evaluate the performances of the ET and GPP models. The advantage of using the T_R-driven ET model under water stress conditions was shown by comparing daily ET from STIC against a widely used non-thermal ET model, called the Priestley-Taylor Jet Propulsion Laboratory (PT-JPL) model (Fisher et al., 2008), in 170 site-years of the 22 flux sites. An overview of the PT-JPL model is presented in Appendix B. Both ET models were implemented with meteorological variables available from the ERA-5 dataset (see Section 2.1.3) and the performances of the two ET models were compared for different dryness levels, as defined in Section 2.1.2. The new GPP model, STIC-RCEEP, and other four LUE models were evaluated and compared for all site-years combined, across site-years of different dryness levels, and for different crop species (C_3 and C_4). In this study, the 170 site-years were grouped into seven dryness levels, namely 0–0.2, 0.2–0.4, 0.4–0.6, 0.6–0.8, 0.8–1.0, 1.0–1.2, and ≥1.2, where the former six intervals are left-closed and right-open. While the lowest level (0–0.2) indicates extreme dryness, the level ≥ 1.2 signifies the wettest limit.

To further understand the potential sources of errors in estimated GPP from STIC-RCEEP, we use a variable named P_{exp}, defined as the proportion of the variations in errors in GPP from STIC-RCEEP (GPP_{err} = estimated GPP – observed GPP) explained by the input variables in a random forest regression (RF) analysis (Breiman, 2001) with 20 random trees across each dryness-level. In this RF analysis, we used GPP_{err} as the target variable and all the inputs of STIC-RCEEP (F_t, EVI, and ET), errors in STIC ET (ET_{err} = estimated ET – observed ET), and three meteorological factors (R_g, VPD, and T_a) as predictors. Further, we define the contribution rates (C_R) of input variables of RF to the explainable variations in GPP_{err} of STIC-RCEEP as $C_R = [cr_1, cr_2, cr_3, \ldots, cr_N]$, where cr_i denotes the contribution rate of variable i with the total number of N variables. We used R^2 computed from the validation subset to determine the value of P_{exp} to avoid overestimation of P_{exp} due to overfitting during the training process. Due to the small size of the validation data, we performed a 5-fold cross-validation, which produced five groups of P_{exp} and C_R values from five different validation and training subsets. Finally, the mean values of P_{exp} or C_R values from the five training-validation processes were used to define the proportion of the explainable variations in GPP_{err} from STIC-RCEEP and the contribution rates of ET_{err} from STIC, all inputs in STIC-RCEEP, and other three meteorological factors (R_g, T_a, and VPD) to the explainable variations.

3. Results

3.1. Evaluation of daily λE from STIC

Evaluations of STIC-derived daily λE against flux site observations and its comparison with another non-thermal ET model, PT-JPL, over 22 cropland sites (170 site-years) are shown in Fig. 2 and the *t*-test results for the difference in performances between the two models are presented in the Appendix (Table D1). Although the analysis over 170 site-years showed similar performance of STIC and PT-JPL (R^2 = 0.62 and RMSE = 26 W m^{-2} from STIC and 0.64 and 26 W m^{-2} from PT-JPL), the performance of STIC was profoundly better under dry conditions (Fig. 2 and Table D1). For example, the cross-site evaluation and comparison between the two models under different dryness categories (Fig. E1) revealed STIC to produce substantially higher R^2 and lower RMSE (mean R^2 (RMSE) = 0.60 (30 W m^{-2})) as compared to the PT-JPL (mean R^2 (RMSE) = 0.47 (36 W m^{-2})) under extremely high dryness (dryness: 0–0.4). The *t*-test analysis further confirmed a significant difference in λE estimates from the two models for all the site-years under high surface dryness (Table D1). Despite PT-JPL captured relatively more variations in λE (R^2: 0.67–0.70) as compared to STIC under the wet conditions (R^2: 0.60–0.64), the RMSE of the two models were comparable when the dryness level is within 0.4 and 1.2. Overall, results indicate that T_R-driven ET from STIC could be more useful in quantifying cropland ET under dry conditions and thus capturing water stress related variations in cropland GPP.

3.2. Parameter optimization of STIC-RCEEP and four LUE models

Optimized values of the undetermined parameters in the five GPP models (Table 4) revealed substantial differences among the four LUE models and the dependency of ε_0 on the GPP model architecture. This finding is consistent with Yuan et al. (2014a), who also revealed different ε_0 values between different LUE models. The values of uWUE•$\sqrt{b}$ (Table 4) was also close to that of Bai et al. (2021), who reported the values of 3272 and 4410 μmol C (mol H_2O) $^{-1}$ kPa$^{0.5}$ for C_3 and C_4 crops, respectively. In this paper, we differentiated crops across the flux sites based on the photosynthesis pathways (i.e., C_3 and C_4 plants) and parameterized ε_0 and uWUE•$\sqrt{b}$ accordingly, which showed higher values for C_4 crops, indicating a greater photosynthetic capacity of this crop type. The ε_0 estimates for the four LUE models for either C_4 or C_3 crops are also different from each other and those for the CASA

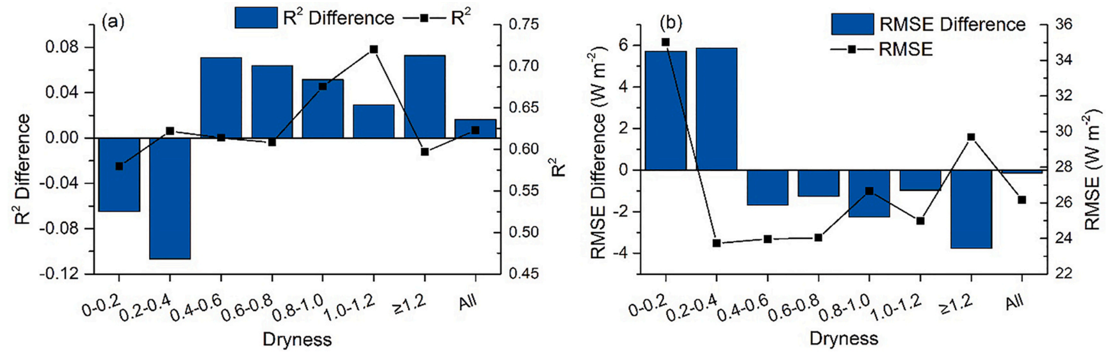

Fig. 2. Performance metric (R^2 and RMSE) differences between PT-JPL and STIC (primary Y-axis) and the performance metric of STIC (secondary Y-axis) in estimating daily λE for site-years across multiple growing season dryness levels. "All" on the horizontal axis indicates the analysis across all dryness levels. R^2 and RMSE were computed for all site-years in each dryness level. R^2 and RMSE differences were calculated as the metrics of PT-JPL minus that of STIC, thus a negative R^2 difference or positive RMSE difference means STIC yielded larger R^2 or smaller RMSE, indicating better performances of STIC as compared to the PT-JPL. The dryness intervals on the horizontal axis are left-closed and right-open.

model, as reported by Chen et al. (2014) (average values of 1.8 gC MJ^{-1} and 2.6 gC MJ^{-1} for C_3 and C_4 plants, respectively). For MOD17, the calibrated value of ε_0 for C_3 crops is about two times greater than the reported value (0.68 gC MJ^{-1} for all crop species) in the original version (Heinsch et al., 2003). This could explain the substantial underestimation of cropland GPP by the original MOD17 model (Zhang et al., 2012; Huang et al., 2018). These comparisons validate the dependency of ε_0 on model architecture. Given our main focus is the validation of the GPP models in croplands, a detailed parameterization of ε_0 or uWUE$\cdot\sqrt{b}$ for different crops is beyond the scope of this study.

3.3. Performance of STIC-RCEEP and GPP model intercomparison

The performances of STIC-RCEEP and four other LUE models using data across different site-years are summarized in Fig. 3 and Fig. E2. Results indicate that the STIC-RCEEP model can reasonably reproduce the variations in crop GPP with R^2 (RMSE) of 0.78 (2.5 gC m^{-2} d^{-1}), 0.72 (2.4 gC m^{-2} d^{-1}), and 0.85 (2.8 gC m^{-2} d^{-1}) for all the data, C_3 crop, and C_4 crop site-days, respectively (Fig. 3a-c). STIC-RCEEP performed significantly better than the other four LUE models (Fig. 3d-f, Fig. E2, and Table 5). Given that STIC-MOD17 and STIC-LUE were also parameterized with STIC ET, and that STIC-LUE typically underperformed among all the models (Fig. 3d-f), STIC-RCEEP was able to

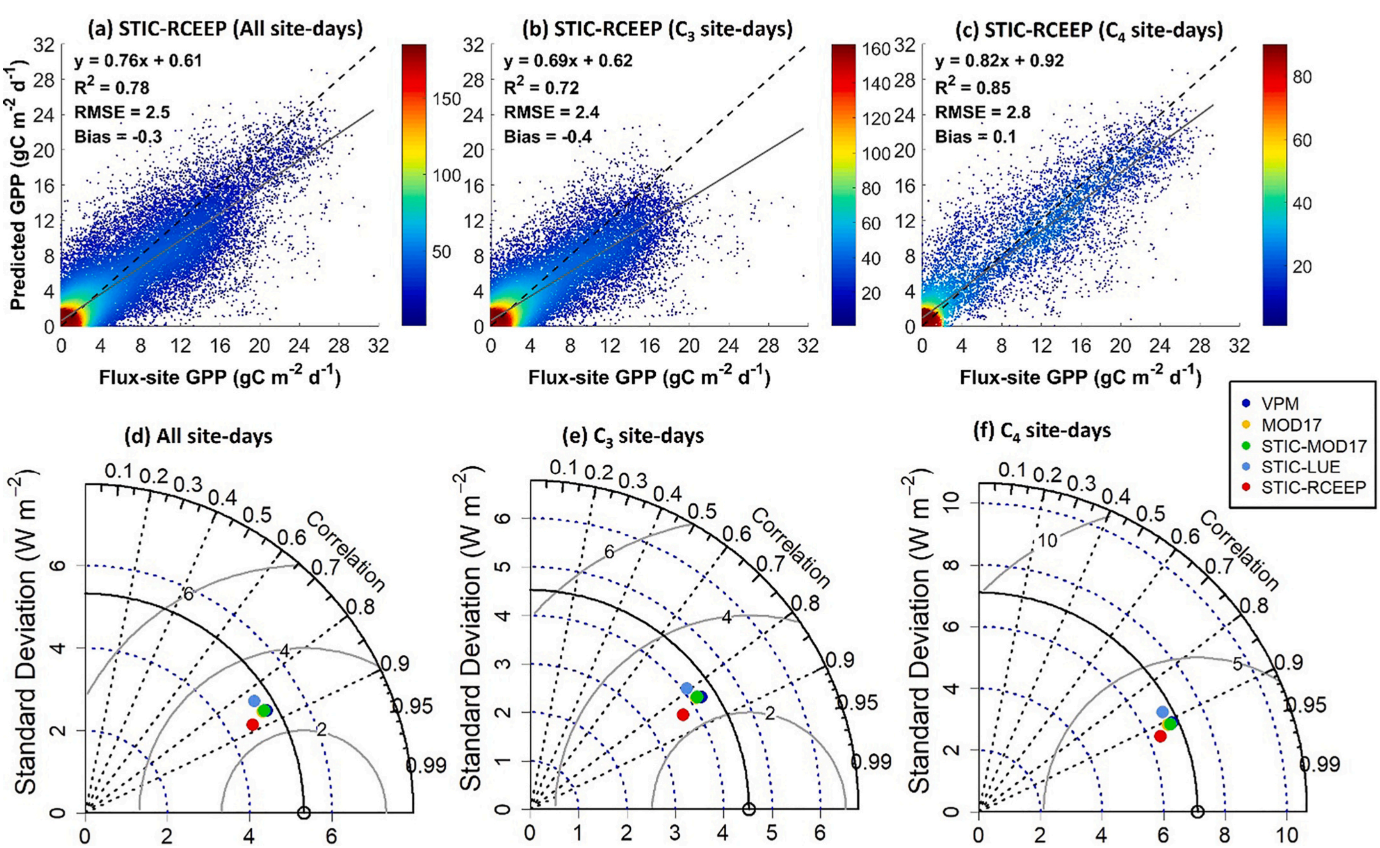

Fig. 3. Cross-site evaluations of daily GPP estimates from STIC-RCEEP for all crop types (a), C_3 (b), and C_4 (c) crops, and an intercomparison with four other LUE models (d), (e), and (f). In the three Taylor diagrams, the colored points having a closer distance to the reference point (empty circle on the horizontal axis) correspond to better performances of the models. The points of VPM, STIC-MOD17, and STIC-LUE are overlapped in (d), (e), and (f).

Table 5
Significance of the difference in performances of STIC-RCEEP with VPM, MOD17, STIC-MOD17, STIC-LUE in estimating daily GPP under multiple dryness conditions. L or R indicates that the model on the left or right performed better with smaller absolute errors (AE) and the symbols that are paired with L or R in the parentheses indicate how significantly better one model was as compared to the other. "-", "*", and "**" represent the *p*-value larger than 0.05, between 0.01 and 0.05, and smaller than 0.01, respectively, indicating nonsignificant, significant, and extremely significant differences in performances between the two models. The *p*-value resulted from the paired-sample *t*-test, which made the null hypothesis that the two models have identical average performance metric values. The dryness intervals below are left-closed and right-open.[a]

Models		Dryness							All
L	R	0–0.2	0.2–0.4	0.4–0.6	0.6–0.8	0.8–1.0	1.0–1.2	≥1.2	
$C_3 + C_4$									
VPM	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	L (**)	L (*)	R (**)
MOD17	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	L (**)	R (–)	R (**)
STIC-MOD17	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	R (–)	R (**)	R (**)
STIC-LUE	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)
C_3									
VPM	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	L (–)	L (–)	R (**)
MOD17	STIC-RCEEP	R (**)	R (**)	R (**)	R (–)	R (**)	L (–)	R (–)	R (**)
STIC-MOD17	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	R (–)	R (**)	R (**)
STIC-LUE	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)	R (**)
C_4									
VPM	STIC-RCEEP	R (*)	R (**)	R (**)	R (**)	R (**)	L (**)	L (*)	R (**)
MOD17	STIC-RCEEP	R (**)	R (–)	L (–)	R (**)	R (–)	L (**)	L (–)	R (**)
STIC-MOD17	STIC-RCEEP	R (–)	R (**)	R (**)	R (**)	R (**)	R (–)	R (**)	R (**)
STIC-LUE	STIC-RCEEP	R (**)	R (**)	R (**)	R (**)	R (**)	L (–)	R (–)	R (**)

[a] "All" denotes the test for site-years across all dryness levels.

better utilize STIC ET to constrain GPP as compared to the other two STIC ET-based LUE models. On the other hand, VPM, MOD17, and STIC-MOD17 showed comparable performances with respect to the correlation and standard deviation between the modeled and *in-situ* GPP. RMSE of daily GPP from STIC-RCEEP was also substantially smaller than those from the other four LUE models (Fig. E2). Overall, the results suggest that STIC-RCEEP can better utilize STIC ET in estimating GPP as compared to STIC-MOD17 and STIC-LUE and perform better than the conventional LUE models for both C_3 and C_4 crops.

3.4. Evaluation of the GPP models under different dryness conditions

The results from the evaluation of STIC-RCEEP and the other four LUE models across the 22 cropland flux sites under different levels of dryness of the site-years are summarized in Fig. 4. Results show acceptable performances of STIC-RCEEP in estimating daily GPP for all dryness levels with $R^2 = 0.76 \pm 0.11$ (data in pairs are "Mean value ± Standard division" and thereafter) and RMSE = 2.7 ± 0.69 gC m^{-2} d^{-1}, better than the other four LUE models. The advantages of STIC-RCEEP were profound under dry conditions (dryness <1.0) (Fig. 4 and Table 5). For instance, for the combination of C_3 and C_4 crops, the R^2 (0.78 ± 0.08) and RMSE (2.5 ± 0.47 gC m^{-2} d^{-1}) of STIC-RCEEP in estimating daily GPP were consistently larger and smaller than those from the other four models for site-years with dryness smaller than 1.0 (Fig. 4).

STIC-RCEEP can also produce acceptable results under wet conditions ($R^2 = 0.77 \pm 0.15$ and RMSE = 2.6 ± 0.25 gC m^{-2} d^{-1}) (Fig. 4). However, there are no notable advantages of STIC-RCEEP when dryness ≥ 1.0, and the model performance appears to degrade with increased surface wetness (Fig. 4 and Table 5). In addition, there were no significant differences between the performance of STIC-RCEEP and those of VPM and MOD17 for C_3 crops under high wetness conditions. Further, under such conditions, VPM performed better than STIC-RCEEP and the other three LUE models for C_4 crops or both C_3 and C_4 crops combined (Table 5 and Table D2). The potential reasons for the relatively weak performance of STIC-RCEEP for wet conditions are discussed in Section 4.3. However, under dry conditions (dryness <1.0), daily GPP estimates from STIC-RCEEP were closer to observed GPP values as compared to those from other models ($R^2 = 0.76 \pm 0.11$ and RMSE = 2.7 ± 0.69 gC m^{-2} d^{-1}). These results revealed notable advantages of STIC-RCEEP under dry conditions and also indicated the potential of further improving the estimates of daily GPP under all conditions, by taking advantage of the abilities of STIC-RCEEP and VPM to capture the water status signals in cropland under dry and wet conditions, respectively.

4. Discussion

4.1. Factors influencing GPP estimates in STIC-RCEEP

As a key input in STIC-RCEEP, any error in λE (λE_{err}) from STIC and other input parameters could be propagated into the estimated GPP. The RF regression analysis showed that approximately 44% of the errors in estimated GPP (GPP_{err}) was explained by λE_{err}, input variables in STIC-RCEEP (P_a, EVI, ET, and f_T), and other factors (R_g, T_a, and VPD) that affect GPP (Fig. 5). While λE_{err} alone and the model inputs explained 10% and 28% of the GPP_{err}, the other factors (R_g, T_a, and VPD) explained only 6% of the errors.

Analyses for different dryness levels showed similar results, with about 42%–47% of GPP_{err} being explained by all variables across the seven dryness levels (Fig. 5), and only a small proportion of GPP_{err} was explained by the other factors. The proportions of GPP_{err} explained by λE_{err} and the model inputs showed large variations across different dryness levels (8%–17% and 20%–30%). The smallest proportions of GPP_{err} explained by λE_{err} was found in the driest site-years (dryness: 0–0.2), while the largest was found in the wettest site-years (dryness: 1.0–1.2). No significant correlation between the proportions of GPP_{err} versus λE_{err} and the dryness was found. Nonetheless, our results indicate that errors in estimated GPP from STIC-RCEEP are primarily associated with the λE_{err} and the input variables, therefore the performance of STIC-RCEEP can be further enhanced by improving the parameterization of the factors that constrain GPP in the model and reducing the errors in STIC λE.

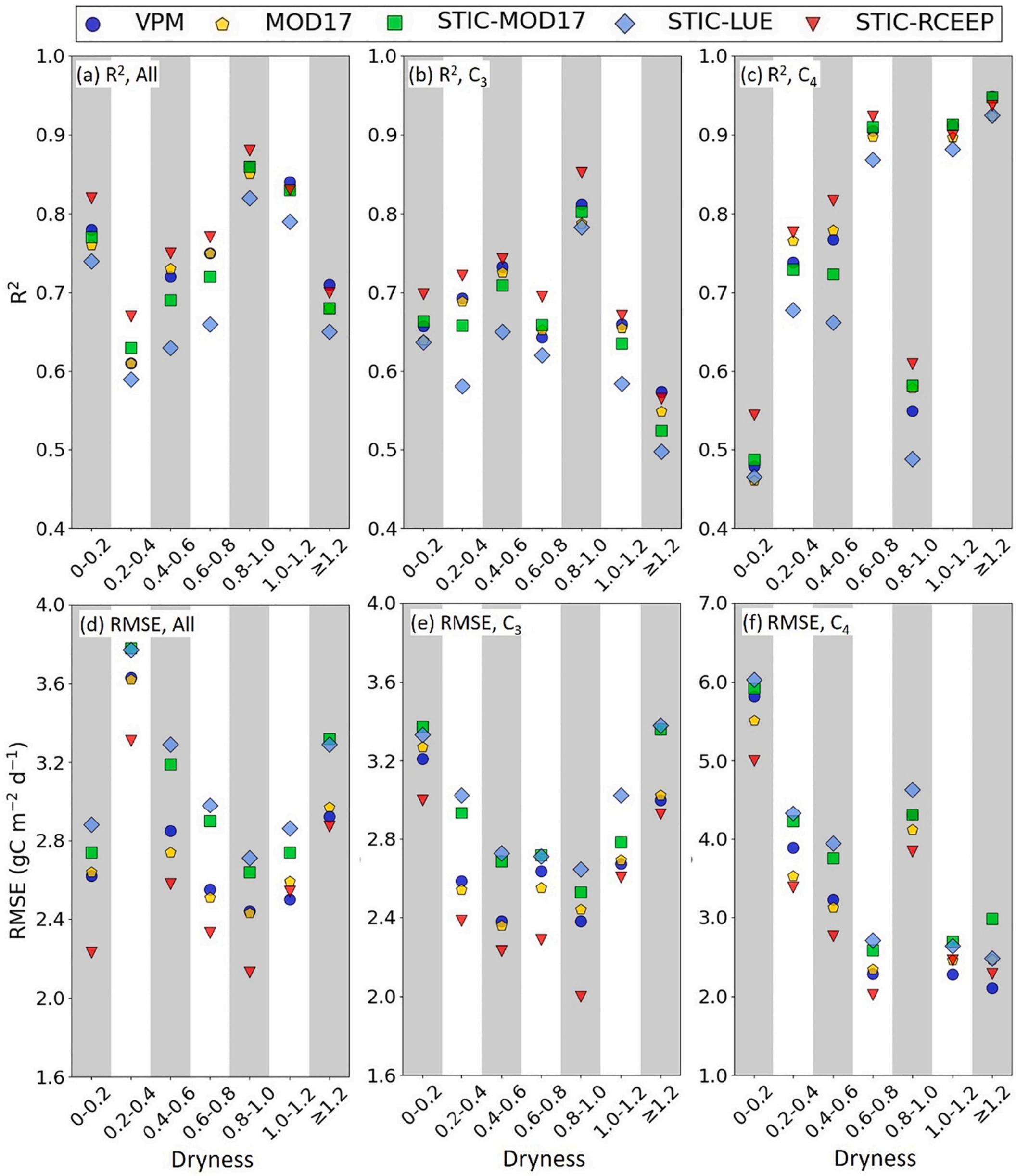

Fig. 4. Comparisons of performance metrics (R^2 and RMSE) between five models for estimating daily GPP in multiple dryness levels. The comparisons were carried out for all site-days (a) and (d), site-days of C_3 crops (b) and (e), and site days of C_4 crops (c) and (f). The dryness intervals on the horizontal axis are left-closed and right-open.

4.2. *Improved parameterizations of ε_0 and uWUE•$\sqrt{b}$ in STIC-RCEEP*

STIC-RCEEP estimates crop GPP using λE from STIC in conjunction with other input variables and results suggest that the model is specifically useful for modeling cropland GPP under dry conditions, compared to traditional LUE models. In addition, STIC-based models can better characterize ε_0 or uWUE•$\sqrt{b}$ by taking into consideration of the photosynthesis pathways of crops (Section 3.2), which highlights a clear advantage of using STIC-RCEEP to estimate GPP as compared to the conventional models. ε_0 is the key factor governing the magnitude and variability of GPP from the LUE models and uWUE•$\sqrt{b}$ functions similarly to STIC-RCEEP. Yuan et al. (2015) revealed significant uncertainties in estimated GPP from the LUE models that do not consider the differences in parameterizations between different photosynthetic pathways. This issue was emphasized by Yuan et al. (2016), in which a constant ε_0 value led to consistent overestimation (underestimation) of GPP from C_3 (C_4) crops. Hence specifying different values of ε_0 or uWUE•$\sqrt{b}$ for C_3 and C_4 crops, as done in this study, is critical to circumvent some of these uncertainties. For example, Chen et al. (2014) suggested that while a dynamic ε_0 among various crop-types within the same photosynthetic pathways is important (due to differences in crop physiology), considering a constant ε_0 value for all crops within a given photosynthesis path, may not result in significant differences in GPP estimates. Yuan et al. (2014a) suggested that although ε_0 can vary between different biomes, the ET-driven LUE models with biome-specified ε_0 and single ε_0 values for all biomes resulted in similar GPP estimates.

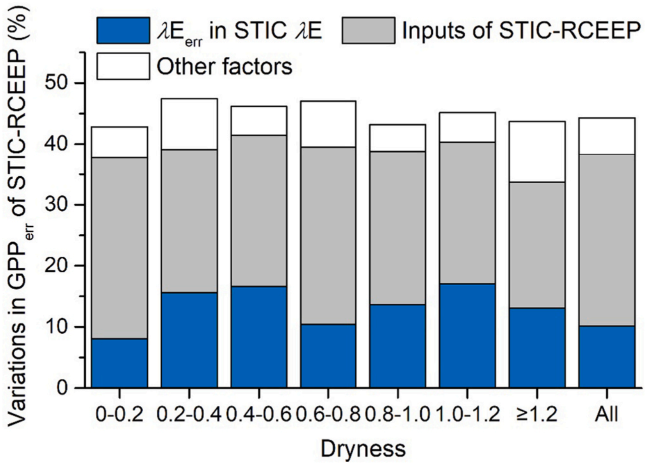

Fig. 5. The variations (%) in GPP_{err} of STIC-RCEEP, explained by λE_{err} of STIC λE, input variables of STIC-RCEEP, and "Other factors" that affect GPP, using the random forest regression method. The "Other factors" are meteorological variables, R_g, VPD, and T_a, which are extensively used to estimate GPP. The dryness intervals on the horizontal axis are left-closed and right-open.

Interestingly, STIC-based GPP models tend to close the gap between ε_0 or uWUE$\cdot\sqrt{b}$ values from C_3 and C_4 crops (Table 4). For example, the - ε_0 values of STIC-MOD17 and STIC-LUE - and uWUE$\cdot\sqrt{b}$ of STIC-RCEEP for C_4 plants were 16%–38% higher as compared to C_3 plants, and these differences are significantly smaller than those of VPM and MOD17 models (47%–57% higher for the C_4 plants) and that reported by Chen et al. (2014) (i.e., 44% higher for C_4 plants). Plant photosynthesis is closely coupled with transpiration. As compared to C_3 plants, C_4 plants have a greater photosynthesis capacity and tend to have larger canopy conductance under low illumination conditions (PAR < 400 μmol m^{-2} s^{-1}) (Huxman and Monson, 2003), and transpire more water under low-humidity conditions (Morison and Gifford, 1983). This explains why the three ET-based models have smaller gaps between ε_0 or uWUE$\cdot\sqrt{b}$ values from C_3 and C_4 crops. This also suggests that STIC-based models can better characterize ε_0 or uWUE$\cdot\sqrt{b}$ by taking into consideration of the photosynthesis pathways of crops, which highlights a clear advantage of using STIC-RCEEP to estimate GPP as compared to the conventional models.

4.3. Sources of error and uncertainties in STIC-RCEEP

Our analyses (Section 4.1) indicate that errors in estimated GPP from STIC-RCEEP are primarily associated with the λE_{err} and the input variables, therefore the performance of STIC-RCEEP can be further enhanced by improving the parameterization of the factors that constrain GPP in the model and reducing the errors in STIC λE. As shown in Section 4.1, errors in the STIC-derived ET can induce substantial uncertainties in the estimates of GPP from STIC-RCEEP. One of the major sources of error in STIC is attributed to the accuracy of satellite-retrieved T_R in MOD11A1 products. Multiple studies reported T_R errors (RMSE) of MOD11A1 products in the range of 0.7–3.1 K with a standard deviation of 1.0 K (Duan et al., 2017; Gomis-Cebolla et al., 2018; Lu et al., 2018). Such errors are primarily associated with surface emissivity and atmospheric water vapor corrections (Wan, 2014; Duan et al., 2017; Lu et al., 2018). MODIS T_R was found to underestimate the true T_R for the barren surface with large positive differences between T_R and air temperature (Li et al., 2014) and the quality of T_R is vulnerable to clouds and wet weather conditions. The uncertainty in the relationship between T_R and the water stress factor (M) is a considerable source of error in the predictive power of STIC as described in Mallick et al. (2018a). A positive (negative) bias in T_R would lead to an underestimation (overestimation) of M and consequent underestimation (overestimation) of g_c as well as overestimation (underestimation) of g_a, respectively. STIC tends to yield larger errors under extremely wet surfaces as indicated by Bhattarai et al. (2018). This explains why STIC-RCEEP lost its advantages over the conventional LUE models that do not use T_R information (Section 3.4). Integrating ET from other T_R-based approaches is beyond the scope of the present study. Bhattarai et al. (2019) have already presented a better performance of STIC compared to six other existing T_R-based ET models, namely METRIC (Allen et al., 2007), SEBAL (Bastiaanssen et al., 1998), SEBS (Su, 2002), TSEB (Kustas and Norman, 1999), Triangular (Jiang and Islam, 2001), and S-SEBI (Roerink et al., 2000), in quantifying basin-scale ET in India.

Errors in STIC-RCEEP could also be because of the effects of spatial resolution of different input variables with respect to the ET and GPP measurements. For example, the T_R data from the MOD11A1 product has a spatial resolution of ~1 km, which is significantly different from the footprint size of ET and GPP measurements at the flux tower scale. MODIS T_R aggregates sub-grid heterogeneity of surface roughness, vegetation index, surface wetness at 1 km^2, whereas most towers aggregate at scales of 0.03–0.2 km^2 for daytime measurements (Chu et al., 2021). This spatial scale mismatch is an important potential source of disagreement between STIC-RCEEP and tower-observed fluxes. Although flux towers are often installed in relatively homogenous terrain, surface wetness and temperatures can still be heterogeneous at the local tower scale, which exerts significant nonlinear effects on ET and GPP (Mallick et al., 2014a). If, the probability of a tower being located in either a cool/wet or hot/dry patch is even, and yet the cool/wet regions contribute disproportionately to the MODIS derived ET and GPP, then, there could be a tendency for the tower-observed flux to be less than its satellite counterpart (Bastiaanssen et al., 1997; Mallick et al., 2014a). Due to the diversity of nonlinear surface characteristics effects on ET and GPP (Mallick et al., 2014a), a detailed evaluation of the scaling characteristics of ET and GPP is beyond the scope of this paper. Given that the flux tower GPP is not observed directly, uncertainties in retrieving GPP from NEE can also lead to additional discrepancies when compared with STIC-RCEEP estimates.

In addition, uncertainties in upscaling instantaneous ET to a daily scale can introduce additional errors (Ryu et al., 2012; Tang et al., 2017). Overall, our results suggested that cropland GPP estimates can be improved under dry conditions using STIC-RCEEP that uses thermal remote sensing derived ET to constrain cropland GPP. This indicates the promise of mapping GPP in arid and semi-arid croplands with the increasing availability of high-quality T_R data.

5. Conclusion

Present study investigated if integrating land surface temperature (T_R) derived ET information in the GPP models can efficiently capture the effects of water stress on photosynthesis and improve cropland GPP estimates under dry conditions. To accomplish this, we incorporated ET estimated from the non-parametric surface energy balance (SEB) model, STIC, with a new coupled model, RCEEP, that estimates GPP from ET. The new model, termed STIC-RCEEP, was evaluated using flux tower-based daily GPP from 170 site-years of 22 cropland flux sites across the globe and its performance was compared against four other LUE-based models. Prior to the use of T_R-driven ET from STIC in GPP modeling, we compared the performance of STIC with another widely used non-thermal ET model, PT-JPL, in terms of estimating daily ET across the studied flux sites. The results are concluded as follows.

(1) The overall assessment of STIC and PT-JPL models across all the site-years and all dryness conditions indicates comparable performances between the two models in estimating daily ET and PT-JPL may be better under the wet conditions. However, under dry conditions, STIC performed profoundly better, suggesting a clear advantage of using the T_R-driven approach in ET and GPP modeling under these conditions.
(2) The optimal ε_0 differs between different architectures of the LUE models as well as the crop photosynthesis pathways. However, the relative difference in ε_0 or uWUE$\sqrt{b}$ between C_3 and C_4 crops can be narrowed by using the T_R-derived ET suggesting the potential of the T_R-based approach in better characterization of these biophysical variables.
(3) The integration of STIC-derived ET in RCEEP showed notable improvement in estimating cropland GPP compared to the LUE-based GPP models. This advantage is more pronounced for the dry site-years (dryness $\leq$1.0) compared to the wet site-years.
(4) Errors in STIC-RCEEP GPP estimates were primarily explained by errors in STIC-derived λE and the inputs variables of STIC-RCEEP, indicating the potential of advancing STIC-RCEEP with improved parameterization of the factors that constrain GPP in the model and reducing the errors in STIC λE.

Overall, we found that the relatively new T_R-based SEB model, STIC, driven by MODIS land surface temperature when integrated with a coupled GPP-ET model can provide a robust means for characterizing cropland GPP under dry conditions. To the best of our knowledge, this study serves as one of the first studies to demonstrate the potential utility of a new T_R-based GPP modeling framework in regional scale GPP mapping over croplands. Finally, our study shows good potential and open avenues for the application of present and future thermal infrared sensors for crop water use and productivity modeling.

Author's contribution

Y.B.: The main conceptual ideas, original manuscript preparation, carried out the experiment, review and editing the manuscript, project administration. *N.B.:* Original manuscript preparation, method conceptualization, review and editing the manuscript, contributing to addressing reviewer's comments, helped with coding the SITC model. *K. M.:* Review and editing the manuscript, modifying the introduction and discussions, writing the STIC part in method section, Section 4.3, and conclusions, contributing to addressing the reviewer's comments. *S.Z.:* Review and editing, processed the experimental data. *T.H.:* Major contributor in helping with STIC code, helped with running the STIC model. *J.Z.:* Review and editing the manuscript, supervision, project administration. All authors have discussed the results and commented on the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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This work used eddy covariance data acquired and shared by the FLUXNET community, AmeriFlux, and European Flux Database Cluster. The FLUXNET also includes these networks: AmeriFlux, AfriFlux, AsiaFlux, CarboAfrica, CarboEuropeIP, CarboItaly, CarboMont, ChinaFlux, Fluxnet-Canada, GreenGrass, ICOS, KoFlux, LBA, NECC, OzFlux-TERN, TCOS-Siberia, and USCCC. The FLUXNET eddy covariance data processing and harmonization was carried out by the ICOS Ecosystem Thematic Center, AmeriFlux Management Project and Fluxdata project of FLUXNET, with the support of CDIAC, and the OzFlux, ChinaFlux and AsiaFlux offices.

Appendix A. The calculations of R_n and G in STIC

A.1. R_n at the satellite overpass time

Net radiation (R_n) at the satellite overpass time for driving STIC was calculated as follows:

$$R_n = R_{g,dir}(1-\text{alb}_{dir}) + R_{g,dif}(1-\text{alb}_{dif}) + R_{nl} \tag{A1}$$

$$R_{nl} = \text{emiss}_s \cdot R_{L\downarrow} - R_{L\uparrow} \tag{A2}$$

$$R_{L\uparrow} = \sigma \cdot \text{emiss}_s \cdot (T_R + 273.15)^4 \tag{A3}$$

$$R_{L\downarrow} = \sigma \cdot \text{emiss}_a \cdot (T_a + 273.15)^4 \tag{A4}$$

$$\text{emiss}_a = 1.24 \cdot (e_a/(T_a + 273.15))^{1/7} \tag{A5}$$

where $R_{g,dir}$, $R_{g,dif}$, R_{nl}, $R_{L\uparrow}$, and $R_{L\downarrow}$ denote the direct shortwave radiation, diffuse shortwave radiation, net longwave radiation, upward longwave radiation, and downward longwave radiation (W m^{-2}), respectively; alb$_{dir}$ and alb$_{dif}$ the surface albedo for direct solar radiation and diffuse radiation; emiss$_s$ and emiss$_a$ the surface and air emissivity; σ the Stefan-Boltzmann constant (5.67×10^{-8} W m^{-2} K^{-4}); T_a and T_R the air temperature and satellite-based radiometric land surface temperature in °C. $R_{g,dir}$ and $R_{g,dif}$ were computed from R_g following Black et al. (1991).

A.2. G at the satellite overpass time and the daily scale

Ground heat flux (G) at the satellite overpass was estimated after Santanello and Friedl (2003):

$$F_G = c_g \cdot cos\left(2\pi\left(t_{g,0} + 10800\right)/t_g\right) \quad (A6)$$

$$G = R_{ns} \cdot F_G \quad (A7)$$

$$c_g = (1 - M_s)c_{g,dry} + M_s \cdot c_{g,wet} \quad (A8)$$

$$t_g = (1 - M_s)t_{g,dry} + M_s \cdot t_{g,wet} \quad (A9)$$

$$t_{g,0} = 43200 - t_0 \quad (A10)$$

where R_{ns} denotes the soil surface net radiation, F_G the fraction of G in R_{ns}; M_s the soil surface moisture availability; c_g is the maximum value of F_G; t_g represents the phase difference between G and R_{ns}; and $t_{g,0}$ is the time in seconds relative to solar noon; t_0 the satellite overpass time in second; $c_{g,dry}$ and $c_{g,wet}$ are c_g values for dry and wet surfaces, and fixed to 0.35 and 0.05, respectively; $t_{g,dry}$ and $t_{g,wet}$ are t_g values for dry and wet surfaces, and fixed to 10,000 and 74,000, respectively.

In this study, F_G on the daily scale ($F_{G,daily}$) was estimated by summing instantaneous F_G weighted by R_n, with the assumption that the variation in R_n from the sunrise (t_{rise}) to sunset (t_{set}) time follows a sinusoidal function:

$$F_{G,daily} = \frac{1}{2}\int_{t_{rise}}^{t_{set}} \left(F_G(t) \cdot sin\left(\pi\frac{t - t_{rise}}{t_{set} - t_{rise}}\right)\right)dt \quad (A11)$$

$$F_G(t) = c_g \cdot cos\left(2\pi\frac{(43200 - t) + 10800}{t_g}\right) = c_g \cdot cos\left(2\pi\frac{54000 - t}{t_g}\right) \quad (A12)$$

where t denotes the local time measured in second; t_{rise} and t_{set} the sunrise and sunset time (in second); and $F_G(t)$ the F_G value at time t. As the numerical solution of $\int_{t_{rise}}^{t_{set}} sin\left(\pi\frac{t-t_{rise}}{t_{set}-t_{rise}}\right)dt$ is 2, the integration on the right side of Eq. (A11) was multiplied by 1/2.

M_s was estimated by decomposing M. Given M is derived from satellite-based T_R, M is composited by the wetness of the soil - that is visible to the satellite - and the wetness of vegetation canopy.

$$M = f_c \cdot M_c + (1 - f_c) \cdot M_s \quad (A13)$$

where f_c denotes the fractional cover of vegetation on a pixel level, and $(1 - f_c)$ indicates the fraction of soil that was visible to the satellite. By inverting Eq. (A13), we can obtain the analytical expressions of M_c and M_s:

$$M_c = \frac{1}{f_c}(M - M_s) + M_s \quad (A14)$$

$$M_s = \frac{1}{1 - f_c}(M - M_c) + M_c \quad (A15)$$

Under the natural environment, vegetation canopies have easier access to solar radiation and larger roughness length as compared to soil, therefore, vegetation drier faster after rain. Based on the above analysis, we decompose M using the following steps:

(a) Initialize M_s by assuming the soil surface was totally wet, namely $M_s = 1.0$;
(b) Compute the wetness of canopy (M_c) using Eq. (A14). If M_c derived in this step is a positive value or zero, stop at this step.
(c) If M_c derived in (b) is smaller than 0, force M_c to be 0 and update the value M_s using Eq. (A15) as well.

Appendix B. The Priestley-Taylor Jet Propulsion Laboratory (PT-JPL)

Priestley-Taylor Jet Propulsion Laboratory (PT-JPL) is a semi-empirical ecophysiological ET model based on the framework of the Priestley-Taylor equation (Priestley and Taylor, 1972; Fisher et al., 2008). It uses visible and near-infrared RS-derived factors and air humidity to estimate the surface water availability for ET rather than the TIR-derived T_R and was found to perform better than most of the existing ET models (Fisher et al., 2020). The model considered the constraints of temperature, the green fraction of canopy, and water availability in vegetation and soil on ET. The key equations of the model are presented as follows:

$$\lambda E = \lambda E_s + \lambda E_c + \lambda E_i \quad (B1)$$

$$\lambda E_c = \alpha(1-f_{wet})f_g f_T f_M R_n f_c \frac{\Delta}{\Delta+\gamma} \tag{B2}$$

$$\lambda E_s = \alpha(f_{wet} + f_{SM}(1-f_{wet}))(R_n(1-f_c) - G)\frac{\Delta}{\Delta+\gamma} \tag{B3}$$

$$\lambda E_c = \alpha f_{wet} R_n f_c \frac{\Delta}{\Delta+\gamma} \tag{B4}$$

where λE_s, λE_c, and λE_i denote the latent heat fluxes equivalent to soil evaporation, canopy transpiration, and canopy interception evaporation, respectively; α the Priestley-Taylor coefficient (1.26); Δ the gradient of the saturated vapor pressure to the air temperature; γ the Psychrometric constant; R_n net radiation; G ground heat flux; f_{wet}, f_g, f_M, and f_c the wet fraction of the surface, the green fraction of the canopy, vegetation moisture, and canopy fractional cover, respectively. Here, G from STIC was used in Eq. (B3); f_c was computed using Eq. (2); and f_{wet}, f_g, and f_M were computed after Fisher et al. (2008).

Appendix C. Dryness of each site-year

Table C1
The dryness of each site-year of 22 flux sites. The symbol "-" indicates that the site-year is not available in this study. Site-years with no more than 100 available daily λE or GPP values were not considered.

Site	Years															
	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
BE-Lon	–	–	–	0.74	0.42	0.83	0.69	1.54	1.04	0.34	0.21	0.54	0.39	0.77	–	–
CH-Oe2	–	–	–	0.99	0.76	1.18	2.05	2.25	1.93	2.54	1.61	2.24	2.66	1.87	–	–
DE-Geb	0.30	0.84	0.43	0.80	0.53	0.48	0.56	0.55	0.79	0.61	0.49	0.55	0.74	0.56	–	–
DE-Kli	–	–	–	0.78	0.62	0.60	0.88	0.53	0.85	0.74	0.71	0.63	2.96	0.72	–	–
DE-RuS	–	–	–	–	–	–	–	–	–	–	0.44	0.65	0.43	0.82	–	–
DE-Seh	–	–	–	–	–	–	0.70	0.75	0.51	0.31	–	–	–	–	–	–
DK-Ris	–	–	–	0.10	0.49	0.66	0.92	0.60	–	–	–	–	–	–	–	–
ES-ES2	–	–	–	0.34	0.11	0.20	0.13	0.22	0.08	0.11	–	–	–	–	–	–
FR-Aur	–	–	–	–	–	0.53	0.65	0.94	0.66	0.65	–	–	–	–	–	–
FR-Gri	–	–	–	0.53	0.41	0.64	0.69	0.53	0.54	0.61	0.50	0.69	0.57	0.65	–	–
FR-Lam	–	–	–	–	0.52	0.46	1.02	0.65	0.60	0.67	–	–	–	–	–	–
IT-BCi	–	–	–	0.60	0.30	0.30	0.29	0.14	0.24	0.32	0.45	1.36	0.45	0.86	–	–
IT-Cas	–	–	–	–	–	0.33	0.30	0.61	0.25	0.56	–	–	–	–	–	–
US-ARM	–	–	0.50	0.82	0.43	0.62	1.48	0.74	0.73	0.69	0.10	0.91	–	–	–	–
US-CRT	–	–	–	–	–	–	–	–	–	–	0.73	0.44	0.58	–	–	–
US-Ne1	1.01	0.92	0.97	0.83	0.98	0.96	1.18	0.90	0.63	1.04	1.17	0.90	–	–	–	–
US-Ne2	1.22	0.74	0.95	0.76	0.96	0.79	1.24	1.15	0.67	0.97	1.08	0.98	–	–	–	–
US-Ne3	0.59	0.45	0.51	0.63	0.57	0.62	0.91	1.14	0.56	0.90	0.90	0.42	–	–	–	–
US-Ro1	–	–	–	–	–	–	–	–	0.60	1.18	0.95	1.03	1.00	–	–	0.77
US-Tw2	–	–	–	–	–	–	–	–	–	–	–	0.05	0.19	–	–	–
US-Tw3	–	–	–	–	–	–	–	–	–	–	–	–	0.12	0.12	–	–
US-Twt	–	–	–	–	–	–	–	–	0.05	0.06	0.11	0.05	0.01	0.02	–	–

Appendix D. Significance test of the difference in performances between models

Table D1
Significance of the difference in performances, as measured using absolute errors (AE) of the modeled values, between STIC and PT-JPL in estimating daily λE under multiple dryness conditions. L or R indicates that the model on the left or right performed better with smaller AE and symbols that are paired with L or R in the parentheses indicate how significant one model was better than the other was. "-", "*", and "**" represent the p-value larger than 0.05, between 0.01 and 0.05, and smaller than 0.01, respectively, indicating nonsignificant, significant, and extremely significant differences in performances between the two models. The p-value resulted from the paired-sample t-test, which made the null hypothesis that the two models have identical average performance metric values. The dryness intervals below are left-closed and right-open.[a]

Models		Dryness							All
L	R	0–0.2	0.2–0.4	0.4–0.6	0.6–0.8	0.8–1.0	1.0–1.2	≥1.2	
STIC	PT-JPL	L (**)	L (**)	R (**)	R (–)	R (**)	R (**)	R (**)	R(–)

[a] "All" denotes the test for site-years across all dryness levels.

Table D2
Significance of the difference in performances, as measured by absolute errors (AE) of the modeled values, between four models (VPM, MOD17, STIC-MOD17, and STIC-LUE) in estimating daily GPP under multiple dryness conditions. L or R indicates that the model on the left or right performed better with smaller AE and the symbols that are paired with L or R in the parentheses indicate how significant one model was better than the other was. "-", "*", and "**" represent the *p*-value larger than 0.05, between 0.01 and 0.05, and smaller than 0.01, respectively, indicating nonsignificant, significant, and extremely significant differences in performances between the two models. The *p*-value resulted from the paired-sample *t*-test, which made the null hypothesis that the two models have identical average performance metric values. The dryness intervals below are left-closed and right-open. [a]

Models		Dryness							All
L	R	0–0.2	0.2–0.4	0.4–0.6	0.6–0.8	0.8–1.0	1.0–1.2	≥1.2	
C3 + C4									
VPM	MOD17	L (–)	R (*)	R (**)	R (**)	R (**)	L (–)	L (**)	R (**)
VPM	STIC-MOD17	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
VPM	STIC-LUE	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
MOD17	STIC-MOD17	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
MOD17	STIC-LUE	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
STIC-MOD17	STIC-LUE	L (**)	R (–)	L (**)	L (**)	R (–)	L (–)	R (**)	L (**)
C3									
VPM	MOD17	L (–)	R (–)	R (**)	R (**)	R (**)	R (–)	L (**)	R (**)
VPM	STIC-MOD17	L (**)	L (**)	L (**)	L (**)	L (**)	L (*)	L (**)	L (**)
VPM	STIC-LUE	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
MOD17	STIC-MOD17	L (**)	L (**)	L (**)	L (**)	L (**)	L (*)	L (**)	L (**)
MOD17	STIC-LUE	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
STIC-MOD17	STIC-LUE	L (–)	R (–)	L (**)	L (**)	R (**)	L (**)	L (–)	L (**)
C4									
VPM	MOD17	L (–)	R (*)	R (**)	R (**)	R (*)	L (–)	L (*)	R (**)
VPM	STIC-MOD17	R (*)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
VPM	STIC-LUE	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
MOD17	STIC-MOD17	R (–)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)	L (**)
MOD17	STIC-LUE	L (–)	L (**)	L (**)	L (**)	L (**)	L (**)	L (–)	L (**)
STIC-MOD17	STIC-LUE	L (**)	L (**)	L (–)	L (–)	L (–)	R (**)	R (**)	R (–)

[a] "All" denotes the test for site-years across all dryness levels.

Appendix E. Figures for comparisons between ET/GPP models

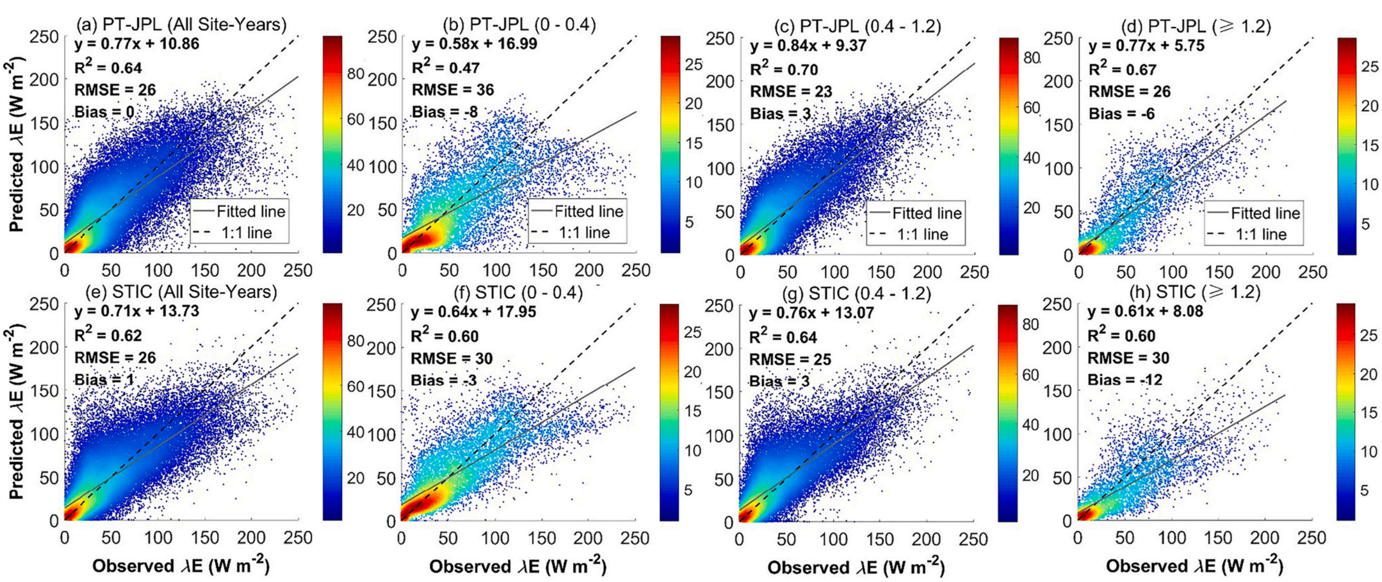

Fig. E1. Estimated latent heat fluxes (λE from PT-JPL and STIC) vs. observed λE across all 170 site-years (a) and (e) and dryness levels of 0–0.4 (b) and (f), 0.4–1.2 (c) and (g), and ≥1.2 (d) and (h). Value ranges in the parentheses indicate the dryness of site-years included in corresponding panels. The color bars indicate the relative density of points within a circular area with a 7.5 W m^{-2} radius. Intervals in the brackets are dryness ranges and are left-closed and right-open.

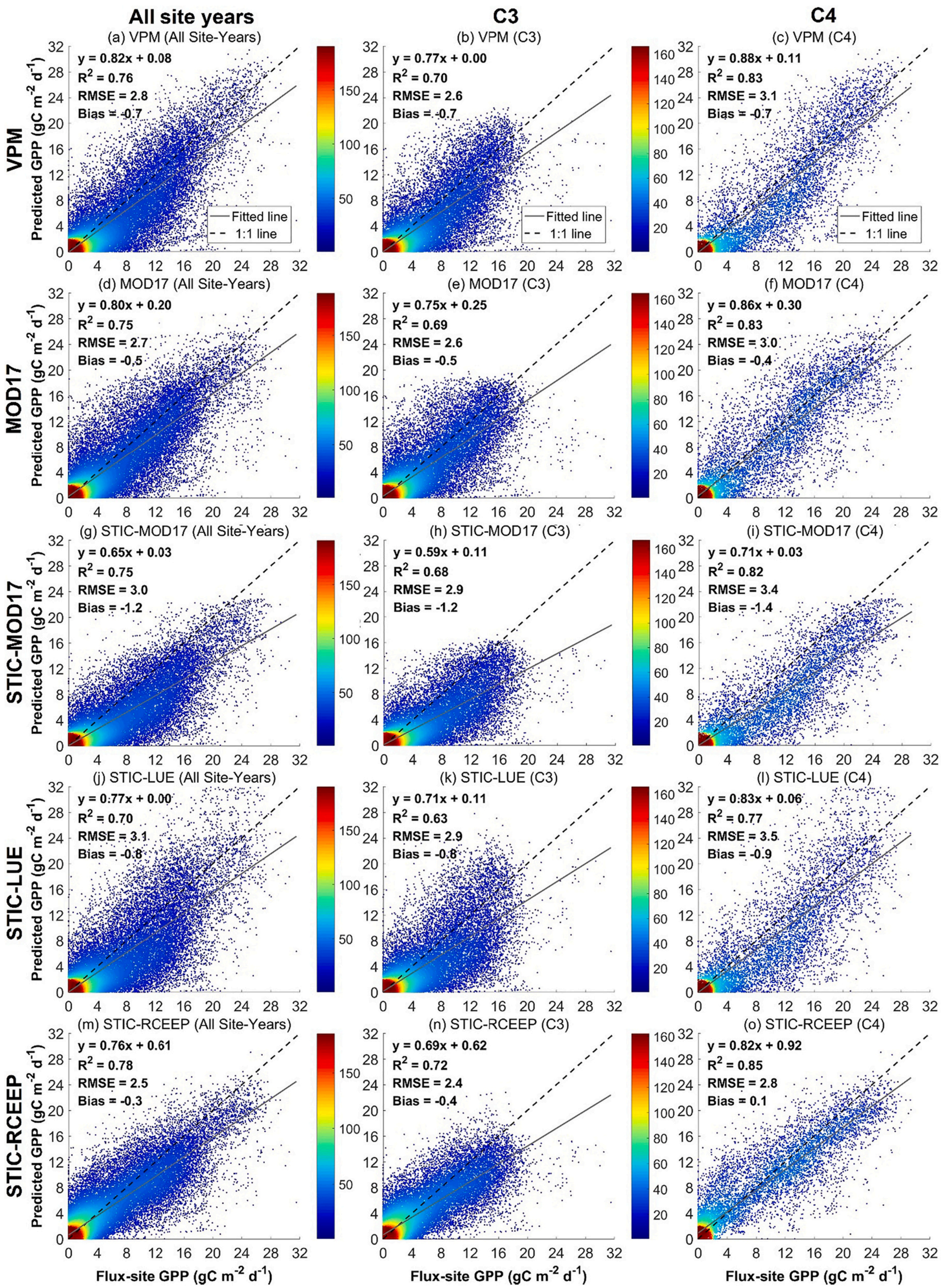

Fig. E2. Evaluations of five models (VPM, MOD17, STIC-MOD17, STIC-LUE, and STIC-RCEEP) to estimate daily gross primary productivity (GPP) for all site-days, site-days for C_3 crops, and site-days for C_4 crops. The color bars indicate the relative density of points within a circular area with a 2.0 gC m^{-2} d^{-1} radius.

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